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SEAT No. _____



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SARDAR PATEL UNIVERSITY

Master of Science (Information Technology) - Semester III Examination - 2022

PS03CINT53 – COMPUTER GRAPHICS & MULTIMEDIA

Wednesday, 09-NOV-2022

TIME: 2.00 pm to 5.00 pm

TOTAL MARKS: [70]

Q.1 Choose the most appropriate option for the below multiple choice questions.

[08]

(i) The maximum number of points that can be displayed without overlap on a CRT is referred to as _____.

- (A) resolution (B) pixel (C) aspect ratio (D) None of these

(ii) DDA stand for _____.

- (A) Digital Differential Analysis (B) Digital Differential Analyzer
(C) Device Differential Analyzer (D) Digital Different Analyzer

(iii) _____ fonts use the frame buffer to store the value for point to be plotted as 1, and others as 0.

- (A) Bitmap (B) Pixmap (C) Outline (D) All of these

(iv) The region against which an object is to be clipped is called a _____.

- (A) clip window (B) clip polygon (C) clip viewpoint (D) clip plane

(v) The photoreceptor cells called _____ are responsible for color identification of the object.

- (A) Fovea (B) Rods (C) Cones (D) Iris

(vi) TIFF stands for _____.

- (A) Total Image File of Form (B) Tag Image File Format
(C) Transcend Image file Format (D) None of these

(vii) The process of removing unwanted sounds that crept in, during the audio recordings is known as _____.

- (A) Trimming (B) Dithering (C) Splicing (D) Noise removal

(viii) Flash is a _____ authoring tool.

- (A) Frame based (B) Icon based (C) Time based (D) All of these

Q.2 Answer the following questions in short. (Attempt any seven)

[14]

(i) Write the applications of Computer Graphics.

(ii) Define the terms :

- (a) Pixel (b) Aspect Ratio

(iii) Assume that a polygon with vertices (2, 4), (12, 4), (2, 10) and (12, 10) is given. Perform the translation transformations to the given polygon with respect to the translation vector (-2, 3). Give the new coordinates of the polygon.

(iv) Discuss Depth Cueing in brief.

(v) Write the steps to convert color values in YIQ graphics format to RGB format.

[P.T.O.]

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- (vi) List out the applications of Digital Image Processing.
- (vii) Write the short note on types of multimedia software.
- (viii) List out various Multimedia applications.
- (ix) Define Hypermedia and Hypertext.

Q.3 Answer the following questions.

- (A) What do you mean by attribute parameter? Discuss attributes of line in detail. [06]
- (B) Discuss DDA line drawing algorithm. [06]

OR

- (B) Discuss Midpoint circle algorithm. [06]

Q.4 Answer the following questions.

- (A) List out the basic two-dimensional geometric transformations. Explain any one in detail. [06]
- (B) What do you mean by clipping? Explain line clipping operation. [06]

OR

- (B) Discuss depth-buffer/Z-buffer method. [06]

Q.5 Answer the following questions.

- (A) Describe the components of Image Processing System. [06]
- (B) Define Image Digitization. Also explain briefly image Digitization process. [06]

OR

- (B) Explain any four Graphics file formats in detail. [06]

Q.6 Answer the following questions.

- (A) Define: Multimedia. Explain in brief the hardware required to handle different media. [06]
- (B) List and explain the components of sound card. [06]

OR

- (B) Write at least 4 reasons for audio editing. Also explain briefly Audio editing terminology. [06]
